The Invariance Principle — problem 1.26

A. Engel — Problem-Solving Strategies · 1998

Three automata I, H, T print pairs of positive integers on tickets. For input (a, b), I and H give $(a+1, b+1)$ and $(a/2, b/2)$, respectively. H accepts only even a, b. T needs two pairs (a, b) and (b, c) as input and yields output (a, c). Starting with $(5, 19)$ can you reach the ticket (a) $(1, 50)$ (b) $(1, 100)$? Initially, we have (a, b), $a < b$. For what n is $(1, n)$ reachable ?